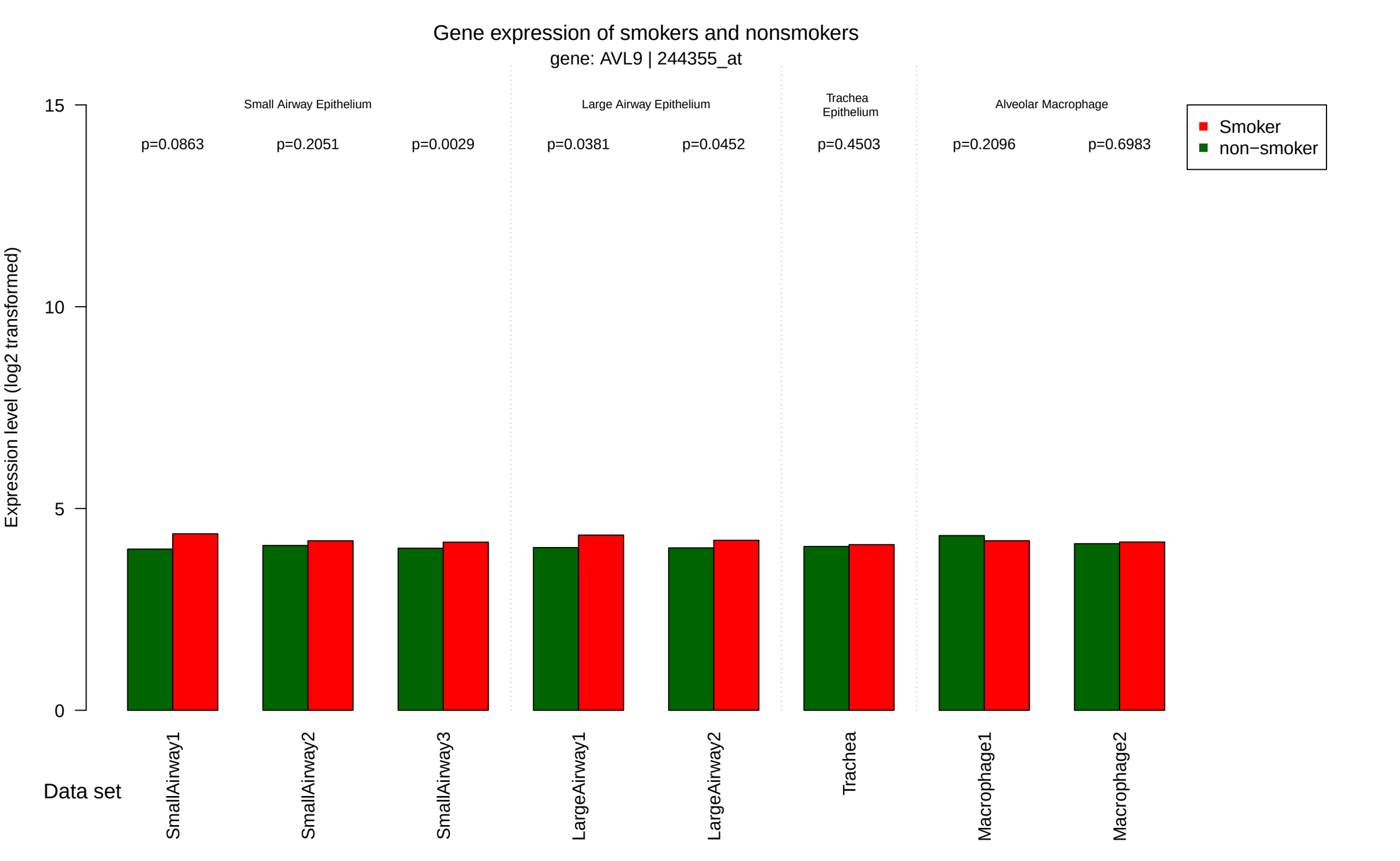
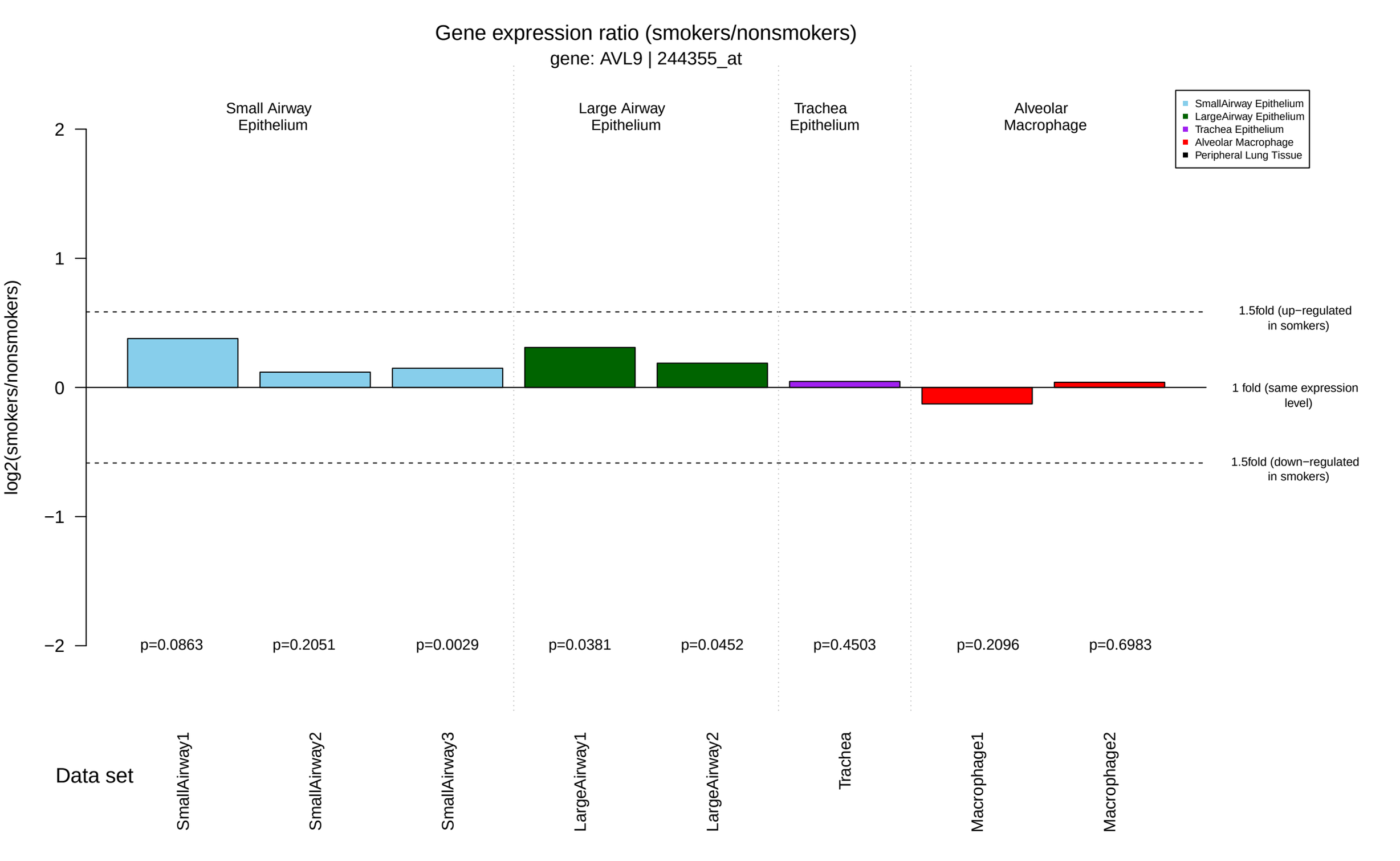
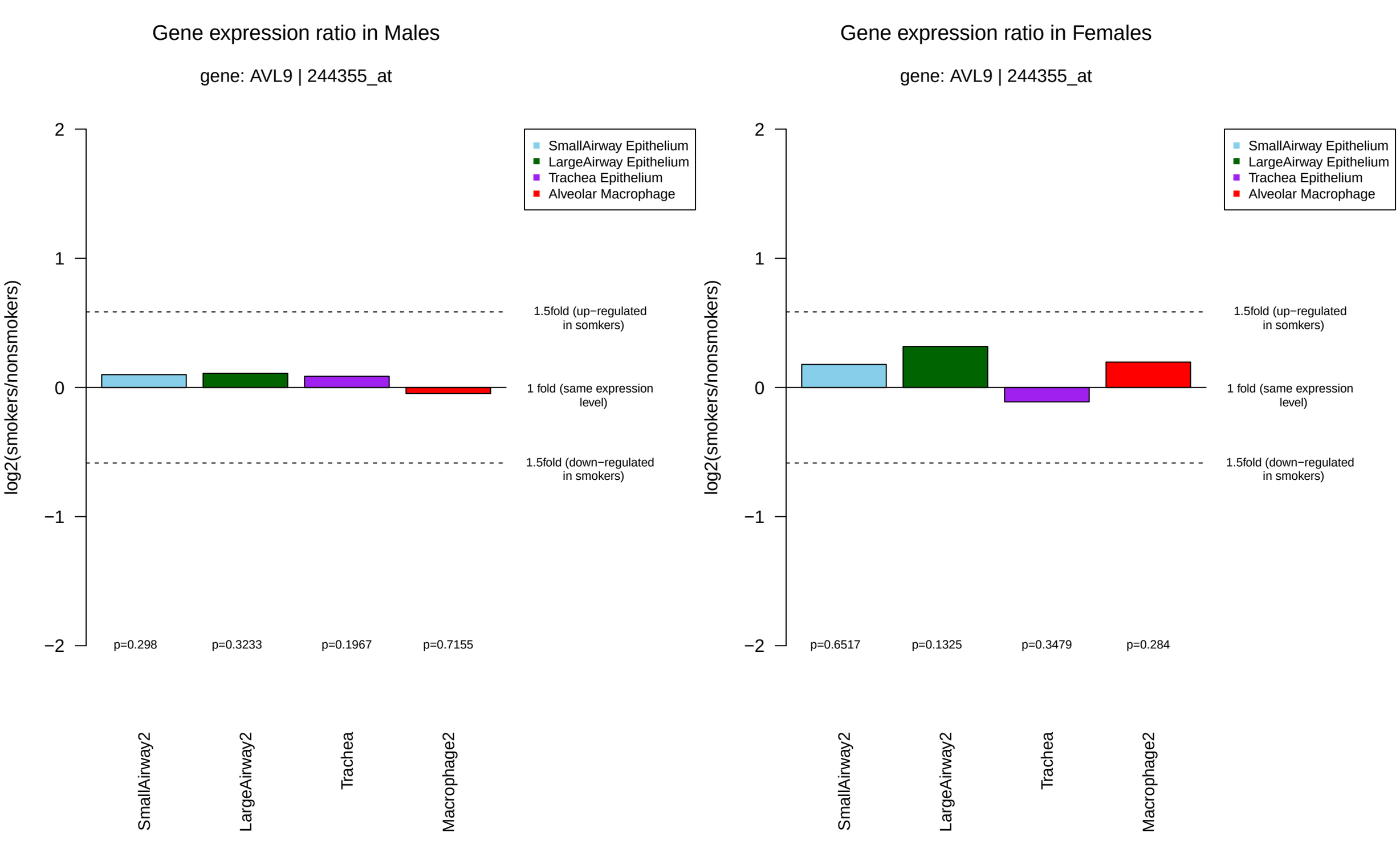
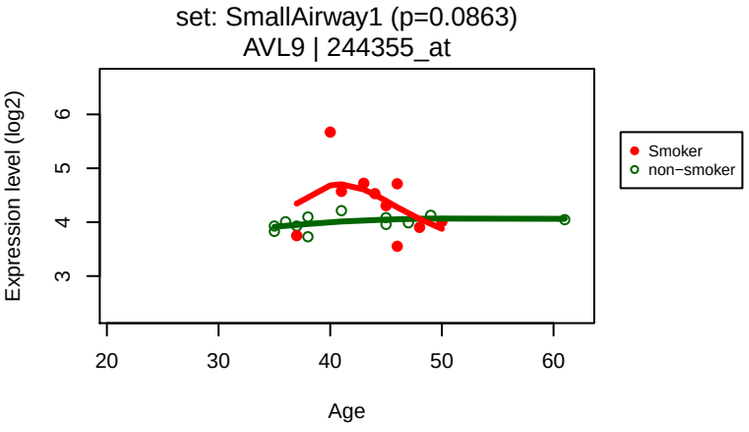
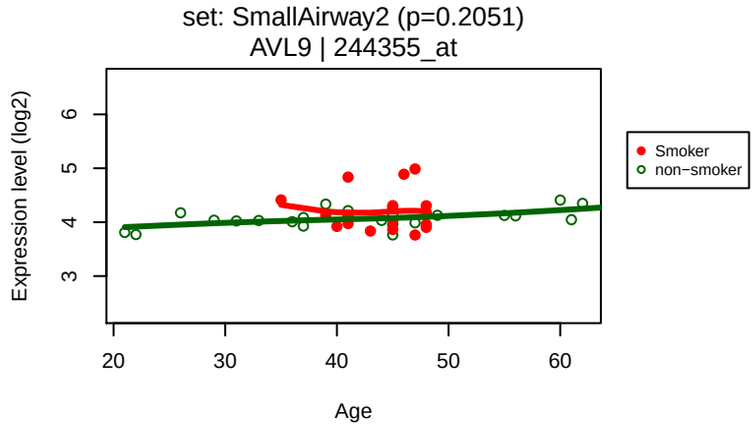
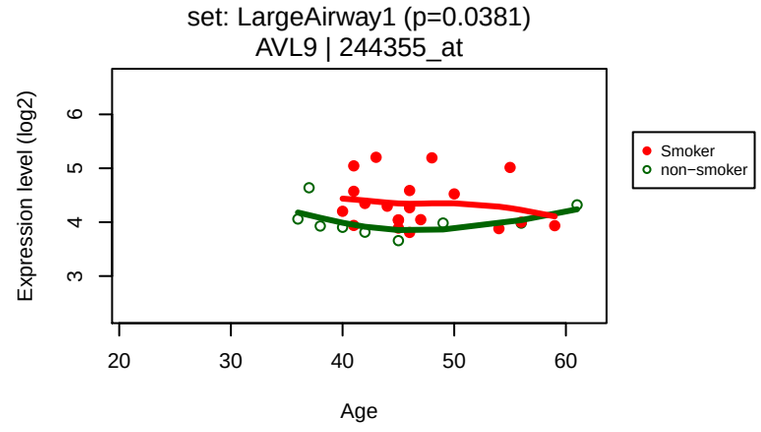

Gene expression of smokers and nonsmokers
gene: AVL9 | 244355_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0863
p=0.2051
p=0.0029
p=0.0381
p=0.0452
p=0.4503
p=0.2096
p=0.6983
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

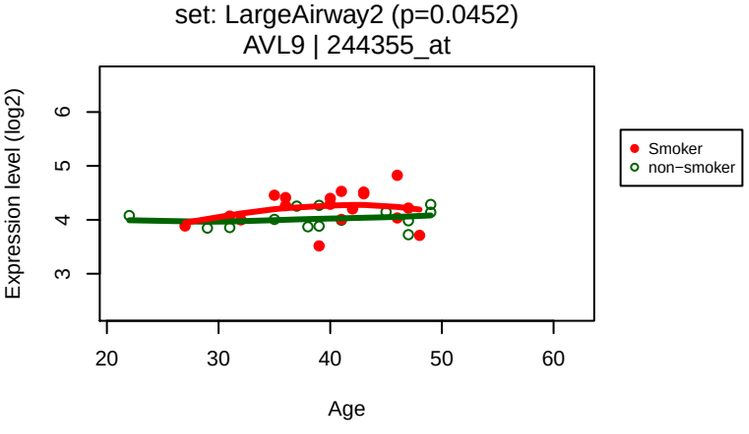

Gene expression ratio (smokers/nonsmokers)
gene: AVL9 | 244355_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0863
p=0.2051
p=0.0029
p=0.0381
p=0.0452
p=0.4503
p=0.2096
p=0.6983
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

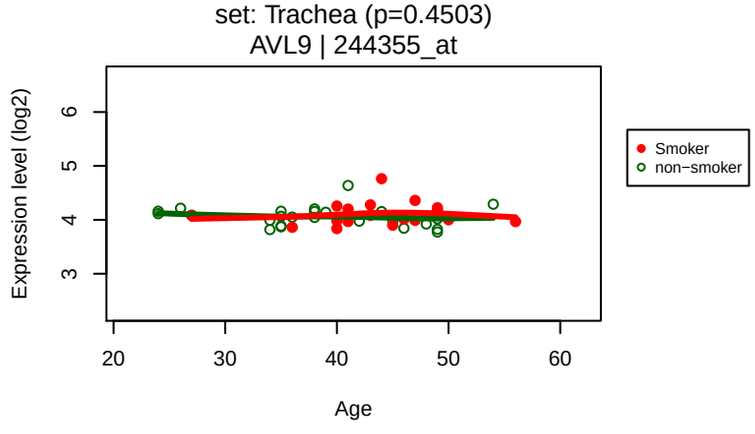

Gene expression ratio in Males
gene: AVL9 | 244355_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.298
p=0.3233
p=0.1967
p=0.7155
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AVL9 | 244355_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6517
p=0.1325
p=0.3479
p=0.284
SmallAirway2
LargeAirway2
Trachea
Macrophage2

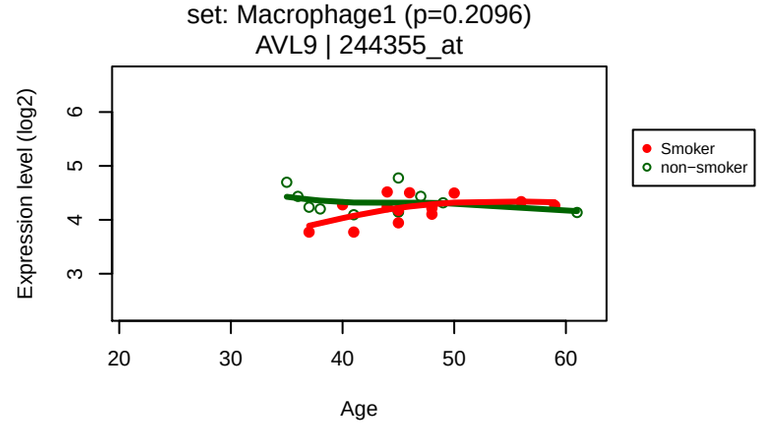

set: SmallAirway1 (p=0.0863)
AVL9 | 244355_at
Expression level (log2)
Age
Smoker
non−smoker

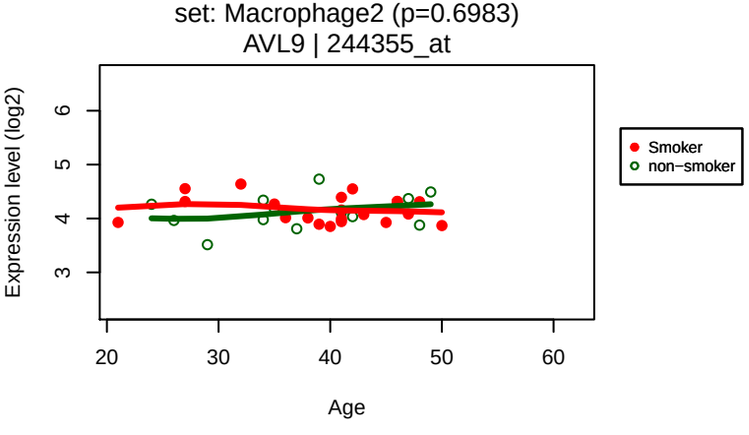

set: SmallAirway2 (p=0.2051)
AVL9 | 244355_at
Expression level (log2)
Age
Smoker
non−smoker

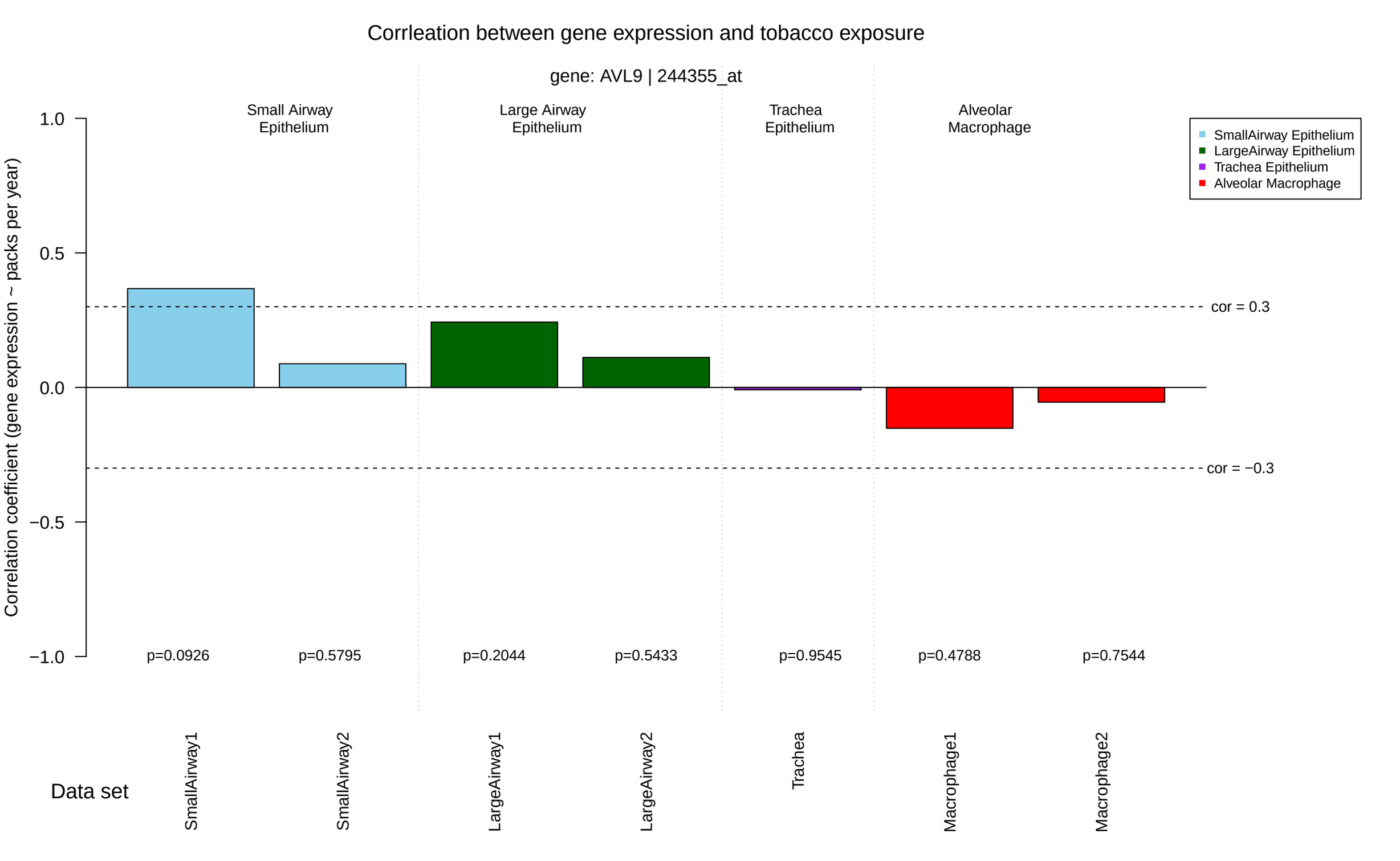

set: LargeAirway1 (p=0.0381)
AVL9 | 244355_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0452)
AVL9 | 244355_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4503)
AVL9 | 244355_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2096)
AVL9 | 244355_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6983)
AVL9 | 244355_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AVL9 | 244355_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0926
p=0.5795
p=0.2044
p=0.5433
p=0.9545
p=0.4788
p=0.7544
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2